

Dynamic Contrastive Learning for Time Series Representation

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 - Representation Learning in Sequential Data
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Presentation overview

Part 1: General overview on self-supervised and contrastive representation learning

Part 2: Dynamic contrastive learning for time series representation (under review)

Current Research: Future work direction

Part 1: General overview on self-supervised and contrastive representation learning

Motivation

- Learning how the brain works and how we learn
- Supervised learning or unsupervised learning

“ The brain's visual system has $O(10^{14})$ neural connections. And you only live for $O(10^9)$ seconds. So it's no use learning one bit per second. You need more like $O(10^5)$ bits per second. And there's only one place you can get that much information: from the input itself ”.
Geoffrey Hinton, 1996



source: university of toronto https://www.instagram.com/uoft/p/DA3lDasx_JM/

How do humans actually learn?



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



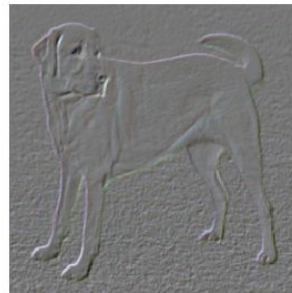
(g) Cutout



(h) Gaussian noise



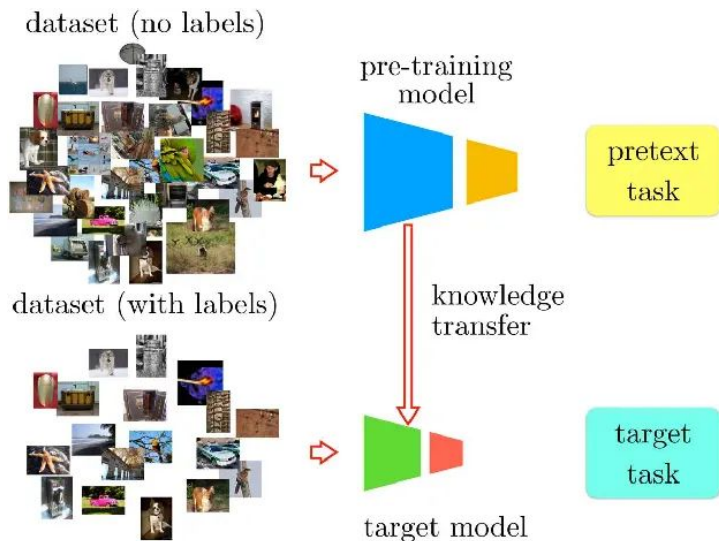
(i) Gaussian blur



(j) Sobel filtering

Self-supervised representation learning

Model trains itself to learn one part of the input from another part of the input through predictive or pretext learning



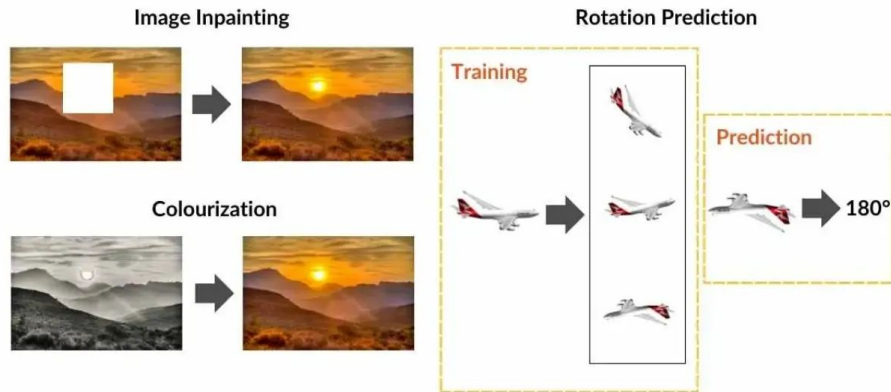
Pretext tasks

- Rotation
- Inpainting
- Colorization
- Jigsaw puzzle
- Reconstruction

Self-supervised representation learning

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Pretext Tasks

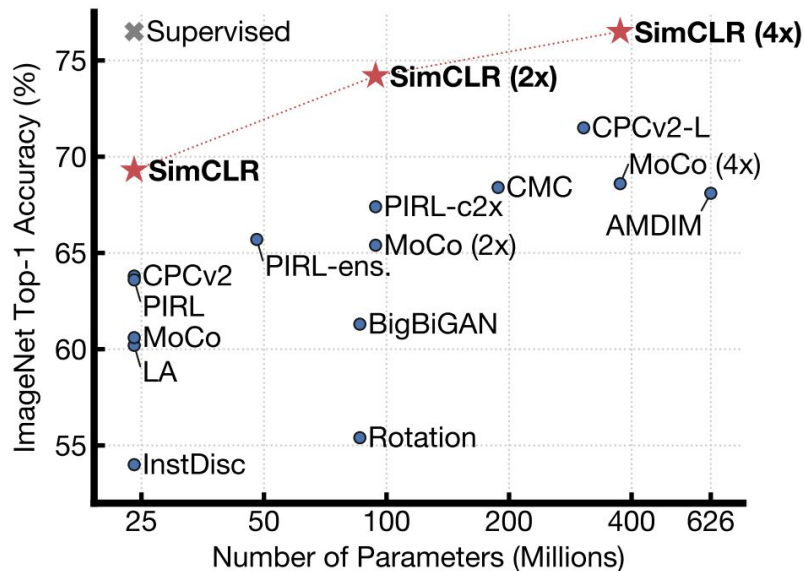


Pretext tasks

- Rotation
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Contrastive representation learning

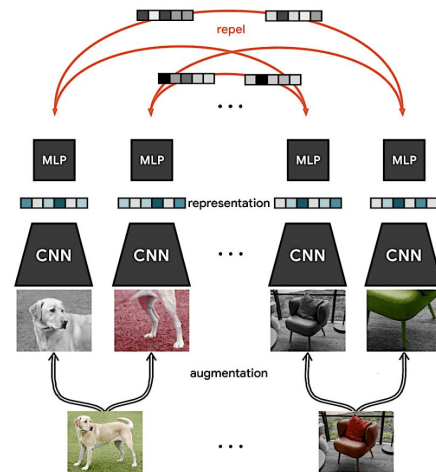
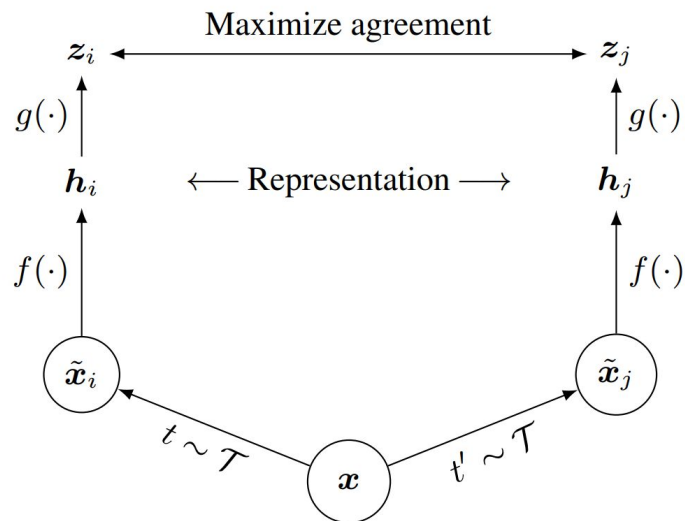
Contrastive-based methods aim to learn data representation by contrasting positive and negative samples



source: <https://arxiv.org/pdf/2002.05709>

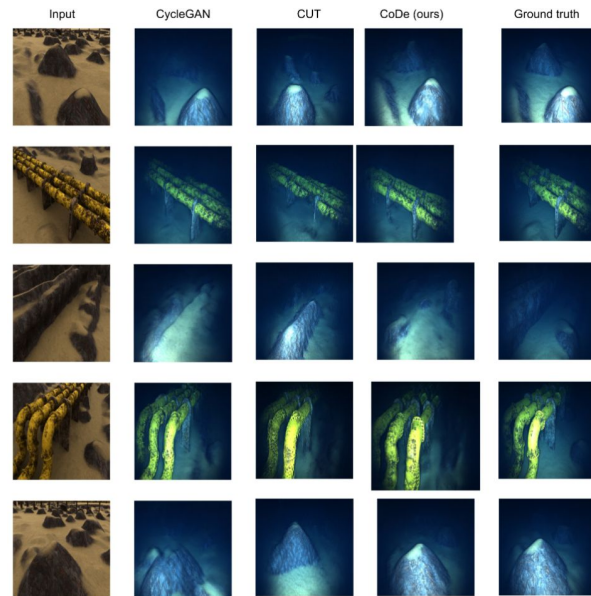
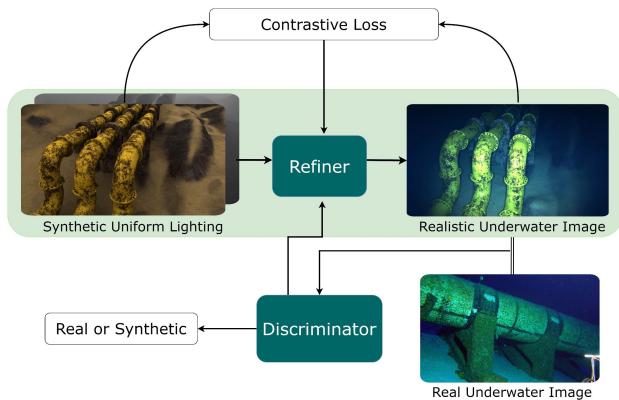
Contrastive representation learning

SimCLR: A simple framework for contrastive learning of visual representations



Contrastive representation learning

Using contrastive learning to enforce similarity between nearby patches in an image



What does augmentations in time series look like?

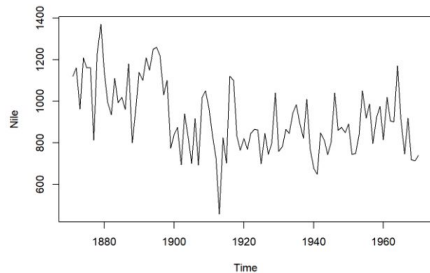
Seeing Through Time: Dynamic Margin Time Contrastive Representation Learning

Abdul-Kazeem Shamba, Kerstin Bach, Gavin Taylor

Image A



Image B



?

image sources: <https://www.sevencycles.com/>, <https://bookdown.org/jqisc/EnvDataSci/its.html>

Looking at the temporal information

Similarity within nearby instances

- That consecutive instances in a sequence have the same class and event labels would not change too often
- Often holds for time series, which have repeated labels in the temporal dimension

Part 2: Dynamic contrastive learning for time series representation (under review)

Contrastive learning in time series

Current state-of-the-arts (SOTA) in time series

CoST (ICLR'22): employs a two-step approach to time series forecasting by learning disentangled seasonal trends

InfoTS (AAAI'23): maximizes agreement between representations of the same subseries through temporal augmentations

TS2Vec (AAAI'22): captures both global and temporal dependencies by contrasting time series across different scales and timesteps

CPC (NeurIPS'18): extracts useful representations by predicting future latent representations in a sequence

TNC (ICLR'21): learns by contrasting data points within the same neighborhood against those from different neighborhoods

Dynamic contrastive learning (DynaCL)

DynaCL adopts N-pair loss to dynamically treat all samples in a batch as positive or negative pairs

DynaCL embeds instances from time series into semantically meaningful clusters

Dynamic contrastive learning (DynaCL)

Key contributions

- Propose DynaCL, an unsupervised contrastive representation learning framework for time series that samples positive pairs from temporal adjacent steps
- Introduce multiple positive pairs in the normalized temperature-scaled cross-entropy loss (NT-Xent) loss to accelerate learning and adapt this for time series representation learning
- Conduct extensive experiments on three public datasets and demonstrate superior results compared to state-of-the-art baselines on clustering and classification

Dynamic contrastive learning (DynaCL)

MP-Xent NT tuple loss per update (efficient training)

$$\ell(i, t) = -\log \frac{\exp(\text{sim}(z_{i,t}, z_{i,t-1})/\tau) + \exp(\text{sim}(z_{i,t}, z_{i,t+1})/\tau)}{\sum_{k=1}^T \mathbf{1}_{[k \neq t, t-1, t+1]} \exp(\text{sim}(z_{i,t}, z_{i,k+1})/\tau) + \sum_{l=1}^T \mathbf{1}_{[l \neq t, t-1]} \exp(\text{sim}(z_{i,t-1}, z_{i,l})/\tau)} \quad (2)$$

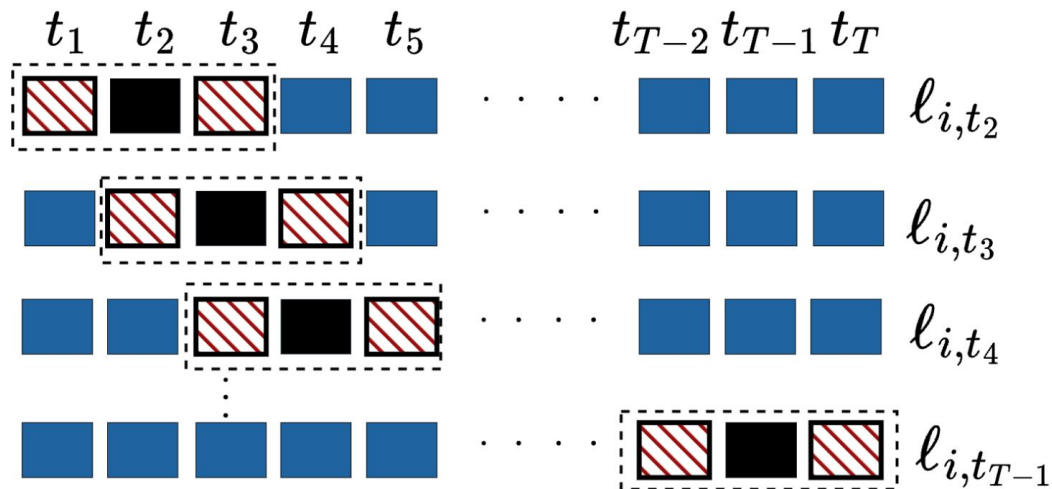
For the entire sequence length T and batch N , we have an NT tuple loss per update, making our training very efficient.

$$L_{MP-Xent} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \ell(i, t) \quad (3)$$

In our DynaCL model, we focus on optimizing only equation [3](#).

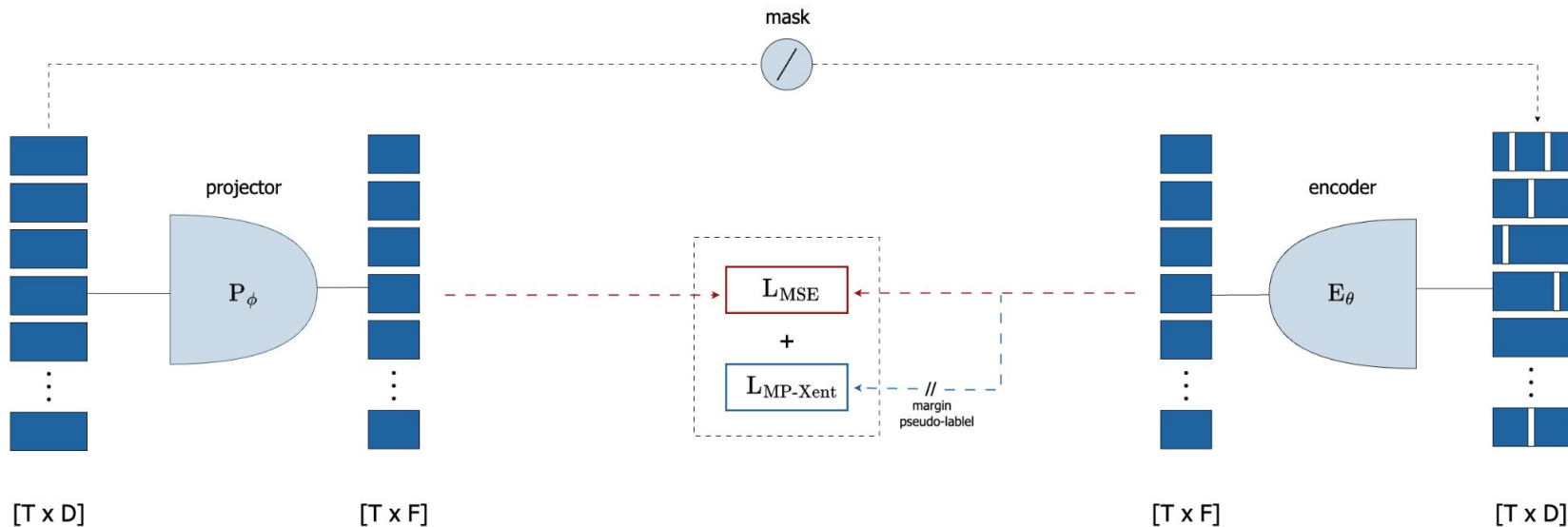
Dynamic contrastive learning (DynaCL)

MP-Xent NT tuple loss per update (efficient training)



Dynamic contrastive learning with margin (DynaCL-M)

DynaCL-M, an augmentation of DynaCL that combines the contrastive learning objective with a masked feature prediction loss



Dataset distributions

Table 1: Summary of dataset distributions used across all experiments. An instance is a preprocessed block of TS. For the HARTH and ECG, each instance is 1 second while for the SLEEP EEG an instance is 2 seconds.

	# Instance	Sequence length	Dimension	Classes	Frequency (Hz)
HARTH	1,270,087	119	156	12	50
SLEEP EEG	371,055	300	178	5	100
ECG	1,531,771	119	252	4	250

Clusterability

Davies-Bouldin Index (DBI) and Silhouette Score

Table 2: Comparison with state-of-the-art methods in clustering. All models are evaluated on test sets that were not used during pretraining. DynaCL-M demonstrates superior clustering performance on two of the three datasets and ranks second to TNC (Tonekaboni et al., 2021) on the ECG.

	HARTH		SLEEPEEG		ECG	
	DBI↓	Silhouette ↑	DBI↓	Silhouette ↑	DBI↓	Silhouette↑
CoST	1.46±0.05	0.25±0.03	2.13±0.18	0.28±0.02	1.15±0.10	0.45±0.01
CPC	1.65±0.11	0.17±0.02	2.59±0.10	0.26±0.01	1.88±0.06	0.16±0.02
TNC	0.60±0.11	0.67±0.24	0.58±0.07	0.21±0.00	0.48±0.06	0.96±0.02
InfoTS	0.96±0.08	0.58±0.03	0.67±0.08	0.24±0.01	0.97±0.10	0.63±0.03
TS2Vec	1.23±0.07	0.59±0.02	1.01±0.05	0.31±0.01	0.74±0.03	0.64±0.02
DynaCL	1.07±0.06	0.35±0.02	1.12±0.25	0.41±0.20	0.98±0.23	0.63±0.13
DynaCL-M	0.46±0.01	0.95±0.01	0.48±0.01	0.87±0.05	0.51±0.11	0.72±0.12

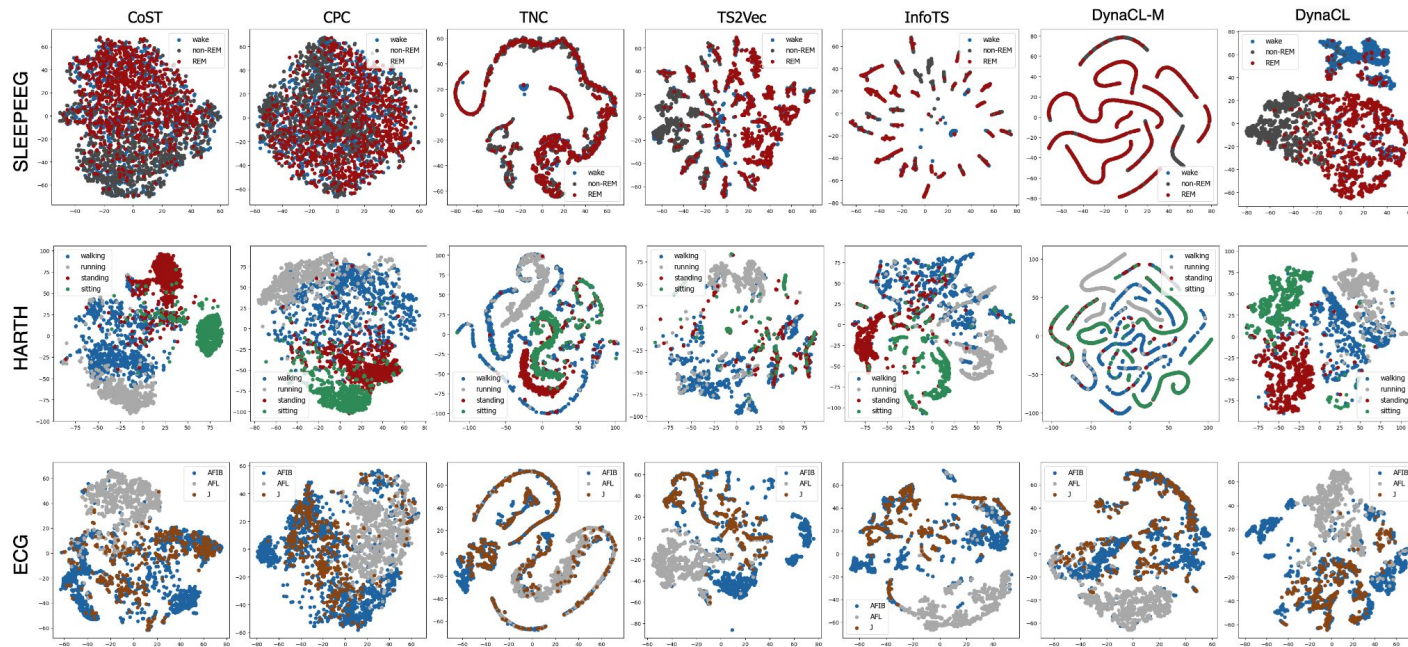
Linear evaluation with frozen backbone

Table 3: Comparison with state-of-the-art methods on linear evaluation with a frozen backbone. We compare DynaCL with state-of-the-art baselines and a randomly initialized encoder (*Random Init.*) on frozen evaluation. We train a linear classifier on top of the from an encoder on the 80% train set (excluding the *all other rhythms* for the ECG dataset) and evaluate the remaining 20%. We train 5 different runs for 50 epochs on all datasets. DynaCL achieved the best performance on all three datasets.

Datasets	Models	Accuracy	F1 score	Precision	Recall
HARTH	Random Init.	34.45±4.01	0.26±0.03	0.18±0.02	0.14±0.01
	CoST	32.74±9.67	0.26±0.06	0.17±0.02	0.15±0.02
	CPC	27.80±4.65	0.20±0.03	0.14±0.01	0.12±0.01
	TNC	30.14±1.04	0.16±0.02	0.06±0.02	0.10±0.01
	InfoTS	33.73±2.48	0.21±0.03	0.16±0.03	0.12±0.02
	TS2Vec	35.58±1.51	0.24±0.02	0.15±0.02	0.12±0.01
	DynaCL	37.95±4.51	0.29±0.06	0.18±0.04	0.13±0.02
	DynaCL-M	31.31±2.73	0.18±0.05	0.08±0.05	0.10±0.01
SLEEP EEG	Random Init.	44.94±0.19	0.32±0.01	0.36±0.01	0.26±0.00
	CoST	50.30±0.23	0.39±0.01	0.32±0.00	0.26±0.01
	CPC	44.19±0.63	0.34±0.02	0.35±0.01	0.26±0.01
	TNC	41.78±0.20	0.26±0.01	0.26±0.01	0.21±0.00
	InfoTS	44.02±0.25	0.34±0.01	0.38±0.02	0.24±0.01
	TS2Vec	48.43±0.49	0.42±0.01	0.42±0.02	0.31±0.01
	DynaCL	62.08±0.64	0.60±0.01	0.52±0.01	0.50±0.01
	DynaCL-M	41.40±4.00	0.28±0.01	0.24±0.01	0.20±0.00
ECG	Random Init.	55.01±0.00	0.51±0.00	0.30±0.00	0.28±0.00
	CoST	55.82±4.95	0.50±0.08	0.31±0.03	0.29±0.02
	CPC	55.36±1.43	0.50±0.04	0.31±0.00	0.29±0.00
	TNC	47.76±0.02	0.31±0.00	0.24±0.01	0.25±0.00
	InfoTS	53.46±0.59	0.45±0.01	0.32±0.01	0.28±0.00
	TS2Vec	49.96±0.65	0.41±0.01	0.27±0.00	0.26±0.00
	DynaCL	58.74±0.62	0.56±0.01	0.32±0.00	0.30±0.00
	DynaCL-M	50.05±0.42	0.37±0.01	0.31±0.00	0.26±0.00

Visualization of learned representations

t-SNE plots of the learned embeddings



Ablation study

Table 4: Ablation study to understand the impact of different components of our model. We notice that only the DynaCL and DynaCL-M variants stand out across all metrics. Clearly, DynaCL-M produces top scores on unsupervised clustering, while DynaCL shows outstanding performance on downstream evaluation.

	HARTH		SLEEP EEG		ECG	
	DBI↓	Silhouette↑	DBI↓	Silhouette↑	DBI↓	Silhouette↑
MP-Xent only (DynaCL)	1.07±0.06	0.35±0.02	1.12±0.25	0.41±0.20	0.98±0.23	0.63±0.13
MP-Xent + MSE	1.05±0.08	0.34±0.03	1.31±0.06	0.24±0.01	0.51±0.11	0.72±0.12
MP-Xent + margin	-	-	-	-	-	-
MP-Xent + MSE + margin (DynaCL-M)	0.46±0.01	0.95±0.01	0.48±0.01	0.87±0.05	0.51±0.11	0.72±0.12

	Linear Acc.	F1 Score	Linear Acc.	F1 Score	Linear Acc.	F1 Score
MP-Xent only (DynaCL)	37.95±4.51	0.29±0.06	62.08±0.64	0.60±0.01	58.74±0.62	0.56±0.01
MP-Xent + MSE	36.27±2.55	0.26±0.04	47.12±0.68	0.39±0.01	50.05±0.42	0.37±0.01
MP-Xent + margin	28.15±0.00	0.12±0.00	41.29±0.01	0.24±0.00	47.76±0.00	0.31±0.00
MP-Xent + MSE + margin (DynaCL-M)	31.31±2.73	0.18±0.05	41.40±0.28	0.28±0.01	50.05±0.42	0.37±0.01

Training time

The simplicity of DynaCL allows it to train very quickly

Table 5: Unsupervised pretraining time (in seconds) for all baseline models over 500 epochs on all three datasets.

	TIME (s)		
	HARTH	SLEEP EEG	ECG
CoST	13.4k	1.5k	3.5k
CPC	4.1k	0.8k	2.1k
TNC	6.5k	1.0k	2.6k
InfoTS	6.1k	1.3k	3.4k
TS2Vec	13.9k	2.3k	5.5k
DynaCL	3.4k	0.6k	1.9k
DynaCL-M	4.6k	1.2k	4.4k

What did we learn?

Key Findings

- Focusing on temporal information rather than augmentation might be the way forward
- Including margin in our MP-Xent objective and jointly optimizing with MSE loss is particularly effective in producing clusters with top scores on the DBI and SS clustering metrics
- Achieving high scores on unsupervised clustering metrics does not necessarily imply that the learned embeddings are meaningful
- Best-performing model for downstream tasks is the vanilla DynaCL without the MSE loss and margin, which proves efficacy of a simple positive sampling strategy of selecting adjacent time steps
- DynaCL model not only delivers exceptional performance in downstream classification tasks but also exhibits the shortest training time.

Future work direction

Anomaly detection using contrastive representation learning ?



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



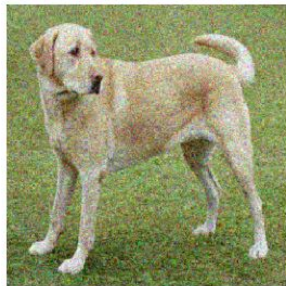
(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering



source: image generated by Microsoft copilot

"We don't need larger models, but better learning algorithms." - NAIIIC'23

Questions?